

SAM-GPT: Brain Lesion Diagnostic Report Generation via Hilbert Curve Enhanced SAM 2 and Vision-Language Models

Anonymous submission

Abstract

Recent breakthroughs in large language models (LLMs) and vision-language models (VLMs) have demonstrated their remarkable capabilities in medical report generation tasks. However, their performance in the specific and challenging domain of brain lesion image report generation remains unexplored. In this paper, we observe that VLMs are prone to misclassification when recognizing brain lesion images. Through feature map analysis, we find that Qwen2.5-VL places excessive attention on the background regions, revealing that the misclassification is caused by backgrounds in medical images that are highly similar to the foreground. Based on this finding, we introduce a novel model, SAM-GPT, which combines a vision foundation model with VLMs for automatically generating brain lesion classification and quantitative results of brain diseases. Specifically, due to the high heterogeneity of brain lesions (e.g., lesioned tissue resembling normal tissue), traditional Mamba scanning approaches, such as VMamba and EVMamba, are prone to overlooking lesions due to their global modeling, which often leads to inaccurate localization. To address this issue, we incorporate a new Hilbert scanning method into Mamba to redesign the encoder and decoder of Segment Anything 2 (SAM 2) for improved lesion attention. Considering that prompts play a critical role in VLMs report generation, we enhance the prompts with lesion information obtained from SAM 2 (e.g., volume size, pixel range, lesion location). We conduct experiments with SAM-GPT on a publicly collected brain lesion dataset that includes gliomas, cerebral infarctions, and epilepsy. The results show that SAM-GPT achieves an average Dice coefficient of 72.8% on the brain lesion segmentation task and an accuracy of 80.25% on the brain disease classification task. Ultimately, SAM-GPT generated a total of 4800 diagnostic reports.

Introduction

Brain lesions profoundly affect patient survival and quality of life. Among brain lesions, cerebral hemorrhage, ischemia, glioma, and epilepsy are the most commonly encountered substantive brain lesions (Hu et al. 2020). Radiologists issue corresponding diagnostic reports based on the patient’s clinical presentation in conjunction with imaging techniques. The automation of brain lesion report generation is crucial for accurate diagnosis and localization of lesions across different imaging modalities. However, writing corresponding radiology reports based on imaging slices involves significant labor and requires experts to possess comprehensive

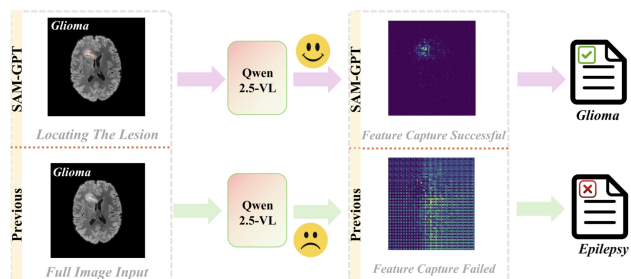


Figure 1: Qwen2.5-VL is prone to interference from image backgrounds, which prevents it from capturing relevant features and leads to incorrect predictions. Once SAM provides Qwen2.5-VL with precise location information, the model successfully extracts the correct features and makes accurate judgments.

diagnostic knowledge. This leads to the accuracy of radiology reports being susceptible to human error and interference due to insufficient physician experience (Awadalla et al. 2023).

CLIP model (Hafner et al. 2021) has provided a new perspective for classification tasks, with some teams training CLIP using matched X-ray image-text data to apply it in the medical field. With the development of VLM models, new approaches have emerged for medical report generation tasks, such as Qwen2.5-VL (Bai et al. 2025) and GPT-4o (Hurst et al. 2024). These VLM models, which are trained on a vast number of image-text pairs from natural scenes, often only require a limited number of medical image-text pairs for fine-tuning to achieve transferability to specific contexts. However, in complex medical scenarios, especially in the presence of lesions, Qwen2.5-VL lacks the capability to accurately identify the location of lesions in images, making it susceptible to interference from the background and potentially leading to misclassifications.

To mitigate the limitations of Qwen2.5-VL in accurately identifying image lesions, we introduce SAM 2 (Ravi et al. 2024) to assist Qwen2.5-VL in lesion localization. However, due to SAM 2’s training being primarily rooted in natural scenes, its direct application in medical lesion localization tends to be suboptimal. Considering the existence of different imaging modalities in medicine that can complement

features, we designed a novel modality interaction approach utilizing the Hilbert curve (Jagadish 1997; Wu et al. 2024). Specifically, in the Encoder component of SAM 2, we re-organized the completed scanning features of Mamba using the Hilbert curve, facilitating efficient interaction between different modalities through the use of an Agent. To ensure the completeness of the decoded features, we replace the original decoder of SAM 2 with a progressively upsampling 3D decoder. Notably, for each 3D decoder, we replaced Self-Attention with the Hilbert Mamba.

In this paper, to assist Qwen 2.5-VL in more accurately recognizing images, we first utilize Hilbert Mamba to aid SAM 2 in mining the correlation of adjacent features based on global modeling characteristics. To leverage imaging information with varying contrasts, we aggregate a new modality interaction mechanism on the basis of Hilbert Mamba to help complete the brain lesion features across images with different contrasts. Finally, we organize the segmentation results from SAM 2 to extract textual information about the lesions, which further enhances Qwen 2.5-VL’s prompts. Specifically, we restructured SAM 2’s Encoder and Decoder using Hilbert Mamba. In SAM 2’s Encoder, we enhanced the sequential inter-expression of adjacent features using the Hilbert Mamba Agent. The enhanced expressive feature sequences from different imaging modalities undergo mutual feature completion through an efficient Agent fusion method, generating the final feature output. In SAM 2’s Decoder, we implemented a 3D Decoder that combines the Memorable Encoder and Memory Bank for progressive upsampling, optimizing the Decoder’s focus on decoding small lesion features. For the 3D Decoder, we replaced the Self-Attention in the Decoder with Hilbert Mamba. Finally, we utilized the lesion volume range, pixel area, and lesion location obtained from the segmentation results of SAM 2, and this textual information further enhanced Qwen 2.5-VL’s input prompts. We conducted experiments with SAM-GPT on a publicly collected brain lesion dataset, which included gliomas, cerebral hemorrhages, cerebral infarctions, and epilepsy. The results show that SAM-GPT achieved a Dice coefficient of 82% for segmenting brain lesions, setting a new state-of-the-art (SOTA). Additionally, the accuracy (ACC) of SAM-GPT in recognizing and correctly classifying the types of brain lesions reached 80%. By combining the segmentation results from SAM 2 with the classification results from LLMs, we generated 4800 reports.

Our main contributions are summarized as follows: (i) We identified a Misclassification issue with Qwen 2.5-VL when recognizing brain lesion images, which we found to be caused by interference from the background of the entire image. For the first time, we attempted to utilize SAM 2 to remove the background from brain lesion images, successfully integrating Vision-Feature Models (VFM) with Large Language Models (LLM). (ii) We restructured the Encoder and Decoder of SAM 2 using Hilbert Mamba, enabling SAM 2 to extract the coupled relationships of local adjacent features while facilitating the interaction of information across different modalities. (iii) We generated corresponding textual information from the segmentation results of SAM 2, for the first time attempting to provide this textual information as

an enhanced prompt to Qwen 2.5-VL.

Preliminaries

Stage Space Models Structured State Space Models (S4) (Gu, Goel, and Ré 2021) and Mamba (Gu and Dao 2023) are sequence models that utilize a hidden state $h(k) \in \mathbb{R}^{N \times 1}$ to map a one-dimensional input sequence $x(k) \in \mathbb{R}$ to an output $y(k) \in \mathbb{R}$, where N indicates the dimensionality of the hidden state. Mathematically, the state space model (SSM) (Chen and Brown 2013) transforms input data through the ordinary differential equations (ODE) $h'(k) = Ah(k) + Bx(k)$ and $y(k) = Ch(k)$, where $A \in \mathbb{R}^{N \times N}$, $B \in \mathbb{R}^{N \times 1}$, and $C \in \mathbb{R}^{1 \times N}$ are the system’s evolution and projection parameters. To facilitate deep learning applications, the continuous system is discretized via the Zero-Order Hold (ZOH) method, resulting in a discrete model, $h_k = Ah_{k-1} + Bx_k$ and $y_k = Ch_k$, where the parameters are transformed as $A = \exp(\Delta A)$ and $B = (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B$. This framework is further leveraged by converting to a convolutional neural network (CNN) form for parallel computation, represented as $y = x \star M$, with M being the convolution kernel derived from the state space model.

Hilbert Curve The Hilbert curve is a type of space-filling curve (SFC) (Moon et al. 2001) that can be represented as $p : [0, 1] \rightarrow [0, 1] \times [0, 1]$, mapping any point in the one-dimensional interval $[0, 1]$ to coordinates in a two-dimensional unit square. We denote the order of the Hilbert curve as n , which approximately corresponds to the height and width of each frame in discrete cases. For any two points u and v in the interval $[0, 1]$, their spatial-to-linear ratio (SLR) is defined as:

$$SLR = \frac{|\sigma(u) - \sigma(v)|^2}{|u - v|}. \quad (1)$$

The expansion factor (DF) of a space-filling curve is defined as the upper bound of the SLR. A lower DF value indicates a tighter mapping in the unit square, aligning with the requirements for preserving locality. Research indicates that the DF of the Hilbert curve is 6, while the DF of the conventional Zigzag curve is given by $4n - 2^{n+1} + 2$, which increases towards infinity as the order n of the curve increases. Therefore, as image resolution improves, the Hilbert curve can more effectively maintain locality when mapping any two points in a one-dimensional sequence to multi-dimensional space compared to the Zigzag curve. To establish global and local correlations, we attempt to incorporate the Hilbert curve into state-space models (SSMs).

Methods

Figure 2 illustrates the overview of our proposed novel two-stage end-to-end architecture, SAM-GPT, for generating reports on substantial brain lesions. Specifically, in the first stage, HilbertSAM outlines the lesion boundaries from the input image sequence $X = \{x_i^m | m = 1, 2, \dots, M; i = 1, 2, \dots, N\}$, generating volume data and spatial location information for the lesions. Based on the lesion information obtained in the first stage, Qwen 2.5-VL utilizes this information to enhance the prompt, classifies the lesions, and

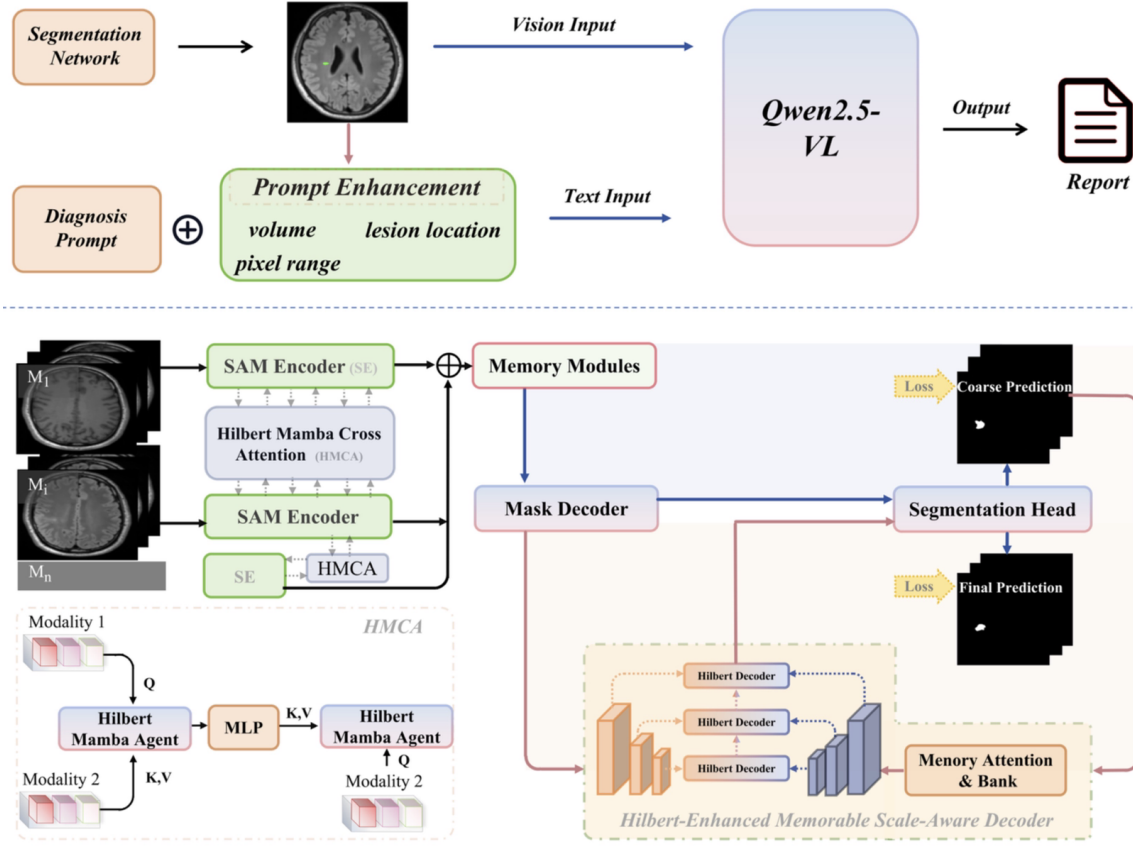


Figure 2: The framework of SAM-GPT consists of two components. The first component utilizes the improved SAM 2 to locate lesions, while the second component leverages the located lesion results as prompts to enhance transmission to Qwen.

combines the lesion information from the first stage with the classification information from the second stage to generate the final diagnostic report.

HilbertSAM primarily consists of a Hilbert-Enhanced Encoder and a Hilbert-Enhanced Decoder. Specifically, the Hilbert-Enhanced Encoder differs from the SAM 2 Encoder by supporting both single-modal and multi-modal inputs. To enhance the recognition capability of small lesions, we modified the SAM 2 Decoder, the Hilbert-Enhanced Decoder effectively leverages features stored in the Memory Bank while incorporating Hilbert Mamba to scan different windows and perform progressive upsampling, ultimately achieving precise localization of lesion spatial positions.

Hilbert Mamba

As shown in the Figure3, we first construct a 3D Hilbert curve that adheres to the principles of Hilbert curves, scanning through every point in the 3D Hilbert space. Essentially, the 3D Hilbert algorithm aims to subdivide the 3D space into multiple cubic subspaces, generating corresponding Hilbert curve segments within each cubic subspace. Finally, these segments formed in the smaller cubic spaces are aggregated to create a complete 3D Hilbert curve. Subsequently, we convert each point on the 3D curve into a unique one-dimensional index. This indexing process flattens the

three-dimensional space according to the scanning trajectory of the Hilbert curve, mathematically represented as:

$$\text{Index} = \hat{x} \cdot Q \cdot T + \hat{y} \cdot T + \hat{z}, \quad (2)$$

where $(\hat{x}, \hat{y}, \hat{z})$ denotes the coordinate index of each point, and Q represents the width length of the input features. Based on the Hilbert index, the features $\hat{F} \in \mathbb{R}^{C \times T \times H/4 \times W/4}$ are flattened into a one-dimensional sequence $\hat{V} \in \mathbb{R}^{C \times (H/4 \times W/4 \times T)}$, which enhances local dependencies compared to the global sequence V . After the local feature enhancement via Hilbert scanning, the sequence is input into the Mamba block, which leverages spatio-temporal corrections in a local context. Ultimately, we reshape the output sequence back to its original dimensions, defined as $\hat{F}' \in \mathbb{R}^{C \times T \times H/4 \times W/4}$. Unlike other scanning methods, the Mamba enhanced with Hilbert scanning places greater emphasis on capturing local dependencies across both spatial and temporal dimensions.

Hilbert Mamba Applications

Hilbert-Enhanced Encoder To assist the Encoder of SAM 2 in implementing the fusion of different imaging modalities, and to enhance its sensitivity in capturing features of small lesions, we have designed a novel Hilbert Agent Cross Attention specifically for the Encoder of SAM

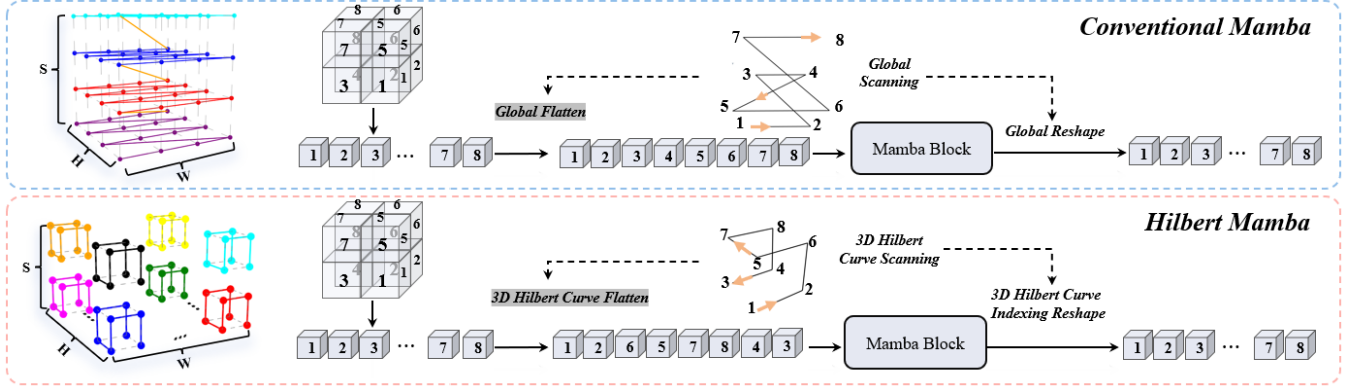


Figure 3: To reinforce the locality-aware scanning of Mamba, we introduce a Hilbert traversal paradigm. The Hilbert curve sequentially threads every element in the space, and its foremost advantage is the exceptional preservation of locality when mapping from one dimension to higher dimensions. This property markedly amplifies feature clustering. By adopting Hilbert scanning, features that reside within the same spatiotemporal neighborhood are densely aggregated along the sequence, thereby strengthening their mutual dependencies.

2, as illustrated in the figure. The Hilbert Agent Cross Attention primarily consists of our newly designed Hilbert Mamba Agent. Specifically, the Hilbert Mamba Agent leverages the properties of Hilbert Mamba to ensure a localized scanning of the feature sequences corresponding to small lesions, while utilizing the Query, Key, and Value provided by different modalities to facilitate the interaction between various imaging modalities.

After the local features are scanned by the Hilbert Mamba, to facilitate interaction between different modalities, the Hilbert Mamba Agent incorporates a novel approach to agent queries. Specifically, given two input modalities m_1 and m_2 , the Hilbert Mamba Agent accepts the Query from m_1 and the Key and Value from m_2 . We introduce a new weight B that replaces the Q values of m_1 . To prevent a significant increase in computational cost due to a large N , the dimension of B differs from that of Q , where B has a dimension of n , and the value of n is much smaller than N . Mathematically, the implementation of the modality can be expressed as:

$$O_1 = \sigma(Qm_1B_1^T + C_2) + \sigma(B_1Km_2^T + C_1)Vm_2 + DWC(Vm_2), \quad (3)$$

where $C_1 \in \mathbb{R}^{n \times N}$, $C_2 \in \mathbb{R}^{N \times n}$, $Qm_1, Km_2, Vm_2 \in \mathbb{R}^{N \times C}$, and $B_1 \in \mathbb{R}^{n \times C}$. C_1 and C_2 are set as learnable parameters for biases. Depthwise convolution (DWC) is introduced to mitigate the issue of insufficient feature diversity in agent attention.

To enhance the deep interaction between different imaging modalities and facilitate continuous low-level feature extraction, after the first Hilbert Mamba, we utilize a Multi-Layer Perceptron (MLP) to extract the first fusion's K and V. We then interact the Query values from m_2 with the extracted K and V values. During the Q, K, V interaction, the Hilbert Mamba scans the newly synthesized sequence again. By continuously scanning local regions, we aim to improve the segmentation of small lesions at the pixel level (e.g.,

small lesions may occupy only 1-10 pixels in the overall pixel space).

Hilbert-Enhanced Decoder Hilbert-Enhanced Decoder is composed of a memory encoder, a memory bank and a Hilbert-Enhanced Scale-Aware decoder. Memory encoder and memory bank create efficient and rich memory representations and store these memory representations. Hilbert-Enhanced Scale-Aware decoder utilizes these representations to optimize the continuity of the three-dimensional feature output. Specifically, to help the decoder decode subtle features from the feature sequence, we attempt to combine a progressive decoding approach with a Hilbert Mamba enhanced 3D decoder. Specifically, we utilize Hilbert Mamba to replace the self-attention mechanism in the original decoder, re-scanning the output tokens and rediscovering features. Unlike previous methods, the Hilbert Mamba applied in the decoder does not perform global scanning during the process; rather, it focuses on reconstructing the relationships of deep features between adjacent tokens. Structurally, the Hilbert-Enhanced Decoder employs three 3D decoders to fuse the pyramid feature maps of the current frame f^t from the mask decoder and the previous frame f^{t-1} from the memory bank at the same resolution through concatenation and convolution. In addition, transposed-convolution is used in the 3D decoder to perform 2x upsampling on the feature maps, which are then fused with upper feature maps. The process is formulated as follows:

$$f_j = \begin{cases} \text{Conv}(\text{Cat}(f_j^t, f_j^{t-1})), & j = \frac{1}{16}, \\ \text{Conv}(\text{Cat}(f_j^t, f_j^{t-1}, \text{UP}(f_{j/2}))), & j \in \{\frac{1}{8}, \frac{1}{4}\}. \end{cases} \quad (4)$$

where $\text{UP}(\cdot)$ represents the transposed-convolution. The output feature maps of the Memorable Scale-Aware Decoder are combined with the feature maps of the mask decoder to enhance the perception capacity of small targets.

Prompt Enhancement

Given that the report generation is based on Qwen 2.5-VL for accurate classification of brain images, we initially employed the Hilbert-Enhanced SAM 2 to precisely delineate the regions of interest to assist Qwen 2.5-VL in accurately recognizing the images. However, considering the Qwen 2.5-VL model’s limitation in capturing critical information related to the volume, lesion boundaries, and lesion locations across layers, we propose a Prompt enhancement method for prompts based on the Qwen 2.5-VL model to address these deficiencies.

Specifically, we define the output of the model without enhancement prompts as $R_{\text{base}} = f(I)$, and the output of the model with enhancement prompts as $R_{\text{enhanced}} = f(I, P)$. We introduce a gain function $G(P)$ to measure the knowledge gain acquired by the model through prompts, which can be expressed as:

$$G(P) = \alpha \cdot \Delta f(I, P) + \beta \cdot \mathcal{H}(I, P), \quad (5)$$

where $\Delta f(I, P)$ is the level of improvement of the model under the enhancement prompts, defined as:

$$\Delta f(I, P) = f(I, P) - f(I). \quad (6)$$

The term $\mathcal{H}(I, P)$ represents the model’s entropy measure for the corresponding input, used to assess the richness of the information. The parameters α and β are weighting factors used to balance the influence of different gains.

We find that the knowledge loss is given by:

$$L_{\text{enhanced}} = \mathbb{E}_{I, P} [L(Y, R_{\text{enhanced}}) | I, P], \quad (7)$$

which can be further expressed as:

$$L(Y, R_{\text{enhanced}}) = \gamma \cdot \|R_{\text{enhanced}} - Y\|^2 + \delta \cdot \mathcal{C}(R_{\text{enhanced}}, Y), \quad (8)$$

where $\|R_{\text{enhanced}} - Y\|^2$ is the mean squared error between the knowledge provided by SAM 2 and the true knowledge. Additionally, $\|R_{\text{enhanced}} - Y\|^2$ represents the mean squared error between the knowledge output by Qwen 2.5-VL and the true knowledge. The hyperparameters γ and δ control the weights of the different loss components.

Consequently, we obtain the overall knowledge gain after enhancement prompts as:

$$R_{\text{enhanced}} = R_{\text{base}} + G(P) \cdot P(Y | I, P), \quad (9)$$

introducing the knowledge loss metric, we express:

$$J_{\text{sa}} = \gamma \cdot \|R_{\text{enhanced}} - Y\|^2 + \delta \cdot \mathcal{C}(R_{\text{enhanced}}, Y) + J_{\text{reg}}, \quad (10)$$

where J_{reg} is a regularization term intended to prevent overfitting.

Experiments

Analysis of Misclassification

To validate the interference of image backgrounds on Qwen 2.5-VL’s ability to discern images, we designed the following experiment: First, we selected 200 medical images (e.g., including 50 T1 images of brain gliomas, 50 FLAIR images of brain gliomas, 50 T1 images of epilepsy, and 50 FLAIR images of epilepsy) and directly fed them to Qwen 2.5-VL

for discrimination. Next, we utilized the unfinetuned SAM 2 to segment these 200 images before passing them to Qwen 2.5-VL for evaluation. Finally, we employed the MedSAM 2 (Zhu, Qi, and Wu 2024), which was fine-tuned on a medical dataset, to segment the 200 images and then provided them to Qwen 2.5-VL for assessment. The experimental results presented in Table 1 demonstrate that the image background significantly interferes with Qwen 2.5-VL’s correct identification of lesions and that effective removal of the image background shows a positive correlation with Qwen 2.5-VL’s accuracy in recognizing lesions.

Datasets

In the task of precise image background removal, we conducted a comparative evaluation of the Hilbert-enhanced SAM 2 network against state-of-the-art methods on three publicly available MRI brain lesion benchmark datasets (e.g., BraTS2021 (Baid et al. 2021), FCD2023 (Schuch et al. 2023), ISLES2022 (Hernandez Petzsche, de la Rosa, and Hanning 2022)) with detailed information available in the Appendix file). We employed metrics such as the Dice coefficient, Intersection over Union (IoU), Precision, and Recall to quantitatively compare the performance of different methods.

For the task of Qwen 2.5-VL in recognizing lesion types, considering that the medical image format cannot be recognized by VLMs, we randomly selected 4,800 2D slices from the constructed brain lesion dataset (e.g., the dataset includes 1,600 slices of gliomas, 1,600 slices of epilepsy, and 1,600 slices of cerebral infarction) and labeled these slices with their corresponding lesion types. We conducted a comparative evaluation of our SAM-GPT network on the 4,800 slices against the latest methods. We utilized ACC, PREC, RECALL, and F1 scores for the quantitative comparison of different methods.

Implementation Details.

The proposed SAM-GPT was implemented using Python 3.10 and PyTorch 2.1.0 on a Ubuntu 22.04 server equipped with ten V100 GPU. In our SAM-GPT model, we configured the initial learning rate to 1×10^{-5} . Throughout training, a cosine annealing scheduler is implemented, which adjusts the learning rate via cosine annealing. A warm-up phase is employed for the first five epochs, during which the learning rate is gradually increased to stabilize the training process.

Method	Acc	Recall	Prec	F1
Qwen 2.5-VL	67%	66%	73%	0.69
SAM 2				
&	61%	64%	72%	0.68
Qwen 2.5-VL				
MedSAM 2				
&	75%	79%	74%	0.76
Qwen 2.5-VL				

Table 1: Validation experiments on Qwen 2.5-VL Misclassifications caused by background.

Method	Year	BraTS2021				FCD2023				ISLES2022			
		Dice(%)	IoU(%)	Prec(%)	Sens(%)	Dice(%)	IoU(%)	Prec(%)	Sens(%)	Dice(%)	IoU(%)	Prec(%)	Sens(%)
UNETR	2022	52.26	48.09	56.75	49.32	26.68	16.09	33.45	25.56	37.27	23.76	39.46	34.32
SwinUNETR	2021	53.27	47.14	54.43	53.33	28.64	16.30	29.71	27.86	37.62	24.06	33.98	41.91
nnFormer	2021	62.36	54.80	63.50	61.45	43.88	30.35	46.91	45.61	50.12	35.21	52.83	46.97
MixUNETR	2025	61.98	52.14	59.65	62.31	38.61	24.52	35.56	41.72	47.74	32.95	49.21	46.39
STUNET	2023	59.80	51.06	60.32	57.11	27.42	16.75	28.81	26.18	22.54	15.82	25.13	28.80
MedSAM	2024	68.44	55.08	69.60	68.21	56.43	40.77	51.20	58.61	66.65	54.27	64.51	71.46
SAM Med 2D	2024	71.38	61.46	73.93	67.35	57.51	42.96	57.30	58.91	67.02	52.16	65.19	69.34
SAM 2	2025	00.03	00.01	00.02	00.03	00.03	00.01	00.01	00.02	00.01	00.01	00.02	00.04
SAM Med 3D	2024	73.56	64.77	75.89	71.33	54.01	37.66	51.09	55.22	69.34	56.40	70.77	67.54
MedSAM 2	2024	76.17	68.43	81.33	75.22	64.22	50.13	62.78	67.01	68.57	57.31	69.65	66.45
FS-MedSAM2	2024	78.84	70.71	79.90	72.11	65.70	50.98	61.01	69.53	65.15	55.07	65.98	64.75
Ours	2025	81.73	72.80	80.29	79.56	66.31	49.83	61.39	73.21	70.38	57.43	71.04	69.59

Table 2: Comparison of segmentation methods on brain lesion datasets. The bolded parts indicate the method with the best current performance for that metric.

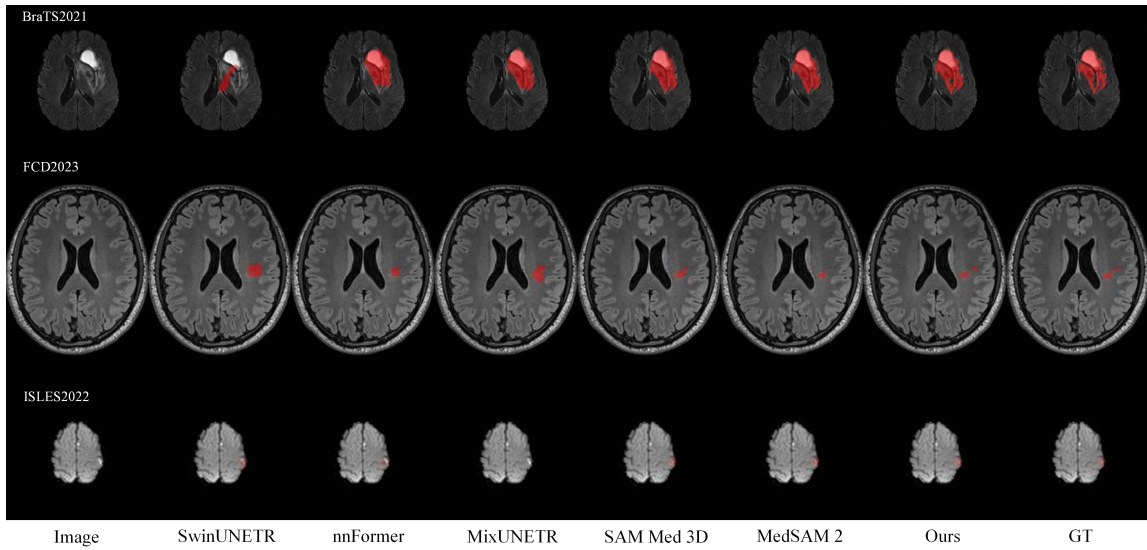


Figure 4: The visualization presentation of segmentation results.

Comparisons with State-of-the-art Methods

Image Background Removal Quantitative Comparisons

To assess the effectiveness of the proposed method on brain lesion datasets, we conducted a comparative evaluation against 11 state-of-the-art methods, including five single-modality 3D segmentation networks (e.g., UNETR(Hatamizadeh et al. 2022), SwinUNETR(Hatamizadeh

Method	Dice	IoU	Params	GFLOPs	Times
SAM 2	00.03	00.01	44	128	156
VMamba+SAM 2	71.56	67.91	102	201	294
EVMamba+SAM 2	75.70	70.22	113	267	322
JamMa+SAM2	73.56	68.84	64	239	229
HM+ SAM 2	81.73	72.80	97	195	178

Table 3: The results of SAM 2 combined with different scanning methods, where HM denotes Hilbert Mamba.

et al. 2021), nnFormer(Zhou et al. 2021), MixUNETR(Shen et al. 2025), STUNET(Huang et al. 2023)), six single-modality segmentation networks based on SAMa(Ma et al. 2024) and SAM 2(e.g., SAM Med 2D(Cheng et al. 2023), SAM 2(Ravi et al. 2024), SAM Med 3D(Wang et al. 2024), MedSAM 2(Zhu, Qi, and Wu 2024), FS-MedSAM 2(Bai et al. 2024)). As shown in the Table2, SAM-GPT achieved the highest Dice coefficient (81.73%), IOU (72.80%), and Recall (79.56%) in the benchmark BraTS2021 dataset tests, indicating its superior capability in brain lesion perception.

To validate the effectiveness of the Hilbert Mamba scanning approach in medical imaging, we integrated Hilbert Mamba with SAM 2 and compared it against other scanning methods—VMamba(Liu et al. 2024), EVMamba(Pei, Huang, and Xu 2025), and JamMa(Lu and Du 2025)—on the BraTS2021 dataset. The evaluation was conducted using metrics including Dice coefficient, IoU, number of parameters, GFLOPs, and running time. As shown in Table3,

Method	HMCA	DHM	PE	ACC(%)	Prec(%)	Recall(%)	F1 Score
M1				61.34	60.74	64.81	0.63
M2	✓			72.73	68.34	74.34	0.71
M3	✓	✓		75.43	71.41	83.47	0.77
SAM-GPT	✓	✓	✓	80.56	83.23	87.30	0.85

Table 4: Quantitative results of our network and the baseline networks constructed from the ablation study (“M1” to “M3”) on 4,800 brain lesion slices.

Method	ACC(%)	Recall(%)	Prec(%)	F1 Score
R2Gen	54.12	53.56	57.36	0.55
R2Gen*	61.48	70.77	60.24	0.65
XrayGPT	60.22	59.45	62.75	0.61
XrayGPT*	64.81	69.60	65.31	0.67
XrayPULSE	61.81	67.42	59.86	0.63
XrayPULSE*	68.73	73.42	67.41	0.70
Med-PaLM M	66.81	65.13	70.27	0.68
Med-PaLM M*	72.63	74.32	69.48	0.72
MiniGPT4	69.33	83.90	64.90	0.73
MiniGPT4*	74.32	80.32	74.73	0.77
Qwen 2.5-VL	67.31	69.79	80.35	0.75
SAM-GPT	80.56	83.23	87.30	0.85

Table 5: Comparison of VLM methods. The asterisk (*) indicates that the SAM-enhanced prompt is re-inputted into the VLM model.

Hilbert Mamba combined with SAM 2 achieves a Dice coefficient of 81.73% and an IoU of 72.09%, outperforming the other scanning methods while maintaining a favorable balance between model complexity and computational efficiency.

Lesion Discrimination Capabilities Comparisons

For the task of lesion discrimination accuracy, we conducted a comparative evaluation of five state-of-the-art methods in the VLM domain (e.g., R2Gen(Wang et al. 2023), XrayGPT(Thawkar et al. 2023), XrayPULSE, Med-PaLM M(Tu et al. 2024), MiniGPT4(Zhu et al. 2023)). As shown in the Table5, SAM-GPT achieved the highest ACC(80.56%), Recall(83.23%), Prec(87.30%), and F1 scores(0.85) in testing 4,800 lesion images. This indicates that SAM-GPT possesses superior capabilities in addressing Misclassification issues. To validate the effectiveness of our proposed solution for misclassification, we re-inputted the outputs of SAM 2 into these five models, accordingly enhancing their prompts. As shown in the Table5, all five models demonstrated a notable performance increase when augmented with inputs from SAM 2.

Visual comparison

The Figure4 visually compares the lesion localization results predicted by our network and state-of-the-art methods on the brain lesion dataset we constructed. Compared to other methods, our network is more effective in localizing lesions.

Ablation Study

Baseline Design We perform ablation experiments to verify the effectiveness of three critical components of our network, namely the Hilbert Mamba Cross Attention (HMCA), the Hilbert Mamba in the Decoder of SAM 2(DHM), and the prompt enhancement from Qwen 2.5-VL(PE). First, we define a baseline model referred to as “M1,” which excludes these three primary components. Next, by incorporating HMCA, we enhance the baseline model “M1,” resulting in model “M2.” Building upon “M2,” we further integrate the Hilbert Mamba from the SAM 2 Decoder to construct “M3.” Finally, we introduce the prompt enhancement from Qwen 2.5-VL. This series of experiments aims to validate the contribution of each component to the overall performance of the network.

Quantitative Comparison The ablation study results presented in the Table4 provide quantitative results of different components of the SAM-GPT method on our constructed dataset of 4,800 brain lesion images. Specifically, compared to “M1,” “M2” improves performance by introducing HMCA, achieving a 11.39% increase in ACC. The addition of DHM in “M3” leads to further improvements, particularly with Recall rising from 74.34% to 83.47%. Finally, incorporating Prompt Enhancement (PE) into the complete SAM-GPT model results in optimal performance, achieving an ACC of 80.56%, a Prec of 83.23%, a Recall of 87.30%, and an F1 Score of 0.85.

Conclusion

In this work, we address the issue of Misclassifications caused by background interference when VLM models recognize images by proposing SAM-GPT, a novel framework that effectively eliminates background distractions and significantly enhances the VLM model’s ability to distinguish lesions. To tackle the challenge of similar appearances between brain lesion tissue and normal tissue, we integrate the Hilbert curve with Mamba and Agent, aiming to explore the deep connections of adjacent features while utilizing multi-modal inputs to supplement feature outputs and improve the model’s localization capability. Additionally, we introduce a Prompt Enhancement technique that leverages lesion volume, pixel regions, and lesion coordinates to assist the VLM model in accumulating prior knowledge about lesions. Comprehensive experiments demonstrate that our method surpasses the current state-of-the-art in both lesion localization and discrimination capabilities.

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